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RESEARCH ARTICLE

Decision analysis of the optimal freight structure at provincial level in China

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Abstract

In recent years, considering the railway and waterway are cleaner ways of freight transportation, the Chinese government has vigorously promoted the adjustment of the transportation structure of the freight industry to reduce carbon emissions. However, with such a macroscopic objective, it remains a crucial problem for the policy-maker to further determine: how to estimate the optimal value of the share of freight volume among roadway, railway, and waterway regionally? To this end, this paper first quantified and analyzed the economic benefits, carbon emission intensity, and input-output efficiency of different transportation modes in each province. Then, with the goal of maximizing the efficiency of the freight transportation system, considering the carbon emission intensity reduction constraint and the economic output non-reduction constraint, an optimization model is constructed to obtain the freight structure adjustment plan in each province. Finally, taking the constraint of the carbon emission reduction targets by the 13th Five-Year Plan period as a case, the specific volume of roadway turnover transferred to waterway and railway in each province is given; the results show that Anhui, Beijing, Jiangxi, Tibet Autonomous Region, and Heilongjiang should focus on transferring the freight share of roadway to the railway, and the transferred shares are 15.36%, 14.12%, 10.21%, 8.22%, and 7.36% of road freight turnover. Tianjin, Hebei, and Shandong should focus on transferring the freight share of the roadway to the waterway, and the transferred shares are 18.44%, 7.90%, and 6.45% of road freight turnover, providing a reference for relevant government departments to formulate measures.

Keywords Freight transport mode · Freight volume share · Greenhouse gas emission · Energy efficiency · Panel model · DEA-SBM model

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Introduction

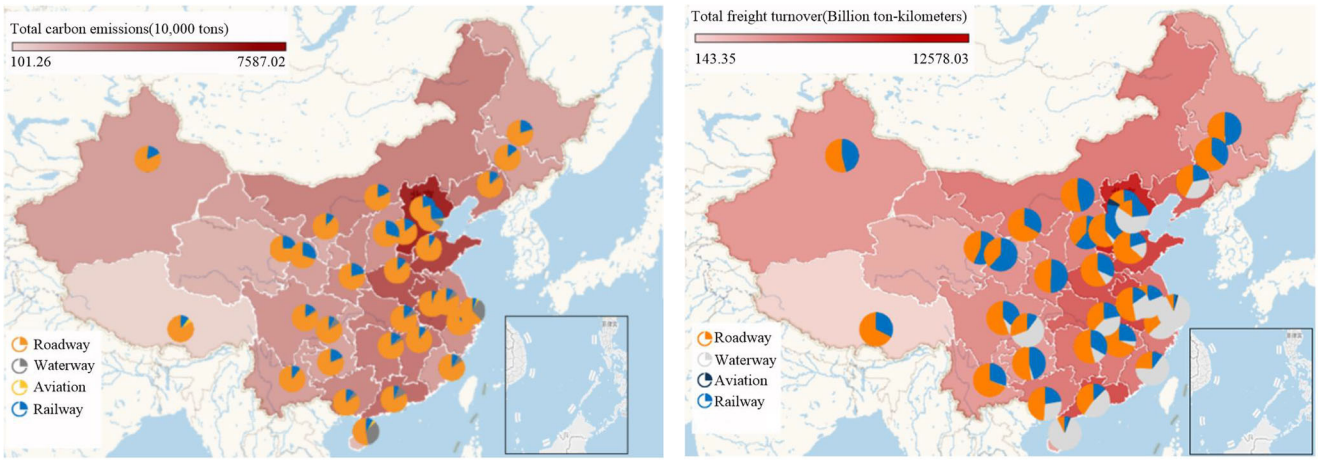
"Low carbon economy" was first seen in government documents in the 2003 British energy white paper "Our energy future-creating a low carbon economy" (UK Department Trade 2003). "Low-carbon economy" aims to build an economic development system with low energy consumption and low pollution, to achieve a win-win situation for economic development and ecological environmental protection. In recent years, sustainable development has become the center of policy discussions in many countries (Sun et al. 2020b). Since 2006, China's carbon emissions have been the highest in the world (Guo et al. 2018). According to the statistics of the International Energy Agency in 2016 (International Energy Agency 2016), the energy consumption in China's transportation sector accounts for more than 30% of total terminal energy consumption in all industries. The greenhouse gases from transportation account for 23% of the total emissions.

Since the China 13th Five-Year Plan, the Chinese government has formulated a series of plans to promote the sustainable low-carbon development of the freight industry (State Council 2014, 2016, 2017). In general, compared with other freight modes, road transportation generates the largest carbon emissions (McKinnon and Piecyk 2009; Li et al. 2013; Wang et al. 2011; Lin et al. 2017; Hao et al. 2015a). Figure 1a shows that among the carbon emissions of different freight transportation modes in 2016 calculated by Formula 11, the carbon emissions of road transportation are the largest in almost all provinces. However, road transportation still undertook the main cargo transportation task, as shown in Fig. 1b. The "Three-Year Action Plan for Promoting the Adjustment of Transportation Structure, 2018-2020" (State Council Office 2018) clearly pointed out that China's existing freight structure is extremely unreasonable and should reduce the share of freight traveling by road while increasing the share traveling by rail, and water. It is worth noting that China has a vast territory, and there are significant differences in the geographic structure, economic level, and energy structure of each province (Jiang et al. 2017; Luo et al. 2016; Jiang et al. 2019; Miao et al. 2019); the impacts of the three freight transportation modes on the economy and carbon emissions also have regional differences (Li et al. 2019). Therefore, the uniqueness of different provinces should also be taken into consideration when formulating a reasonable freight structure adjustment plan.

In recent years, many scholars began to set their sights on the impact of transport modes on the freight industry's carbon emissions. Li et al. (2013) estimated the carbon emissions from road freight transportation in China from 1985 to 2007, and suggested that freight mode shift policies can reduce roadway carbon emissions. Tian et al. (2014) examined the freight greenhouse gas (GHG) emissions of different provinces in China; the results showed that there are significant differences in the carbon emissions intensity of the five transport modes and different provinces, and recommended that a strategy for structural adjustment of transportation modes should be formulated according to the realities of different regions in China. Hao et al. (2015a) established a set of scenarios to reflect the possible future trend of energy consumption and GHG emissions from China's freight transport sector, suggesting to promote a mode of shift from road to rail. Similarly, in many other studies, truck-to-railway and truck-to-waterway in the freight industry are considered to be better choices for the development of green transportation (Luo et al. 2016; Li and Zhang 2020; Lin et al. 2017; Gan et al. 2020). Sim (2017) developed a system dynamics model to estimate the total carbon emissions of South Korea's truck sector. And through their established mathematical model, they calculated the optimal transportation volume of light trucks, medium trucks, heavy trucks, and railway transportation while meeting the anticipated transportation volume and carbon emission quotas.

Most studies have revealed the carbon emission intensity of different freight modes, and have qualitatively given the freight volume adjustment suggestions. A few studies have studied the proportion of different freight modes under the carbon emission limit. However, as far as we know, there is almost no research that has given a clear freight structure adjustment plan to meet China's emission reduction targets and improving the economic efficiency of the freight transportation system at the provincial level.

This paper quantified and analyzed the economic benefits, carbon emission intensity, and input-output efficiency of different transportation modes in each province. First, the panel data model was established to calculate the output value per unit freight turnover volume of roadway, railway, and waterway. And the top-down carbon emission calculation formula was used to calculate carbon emission of three transportation modes. The Data Envelopment Analysis and Slacks-Based Measure (DEA-SBM) model evaluated the efficiency of the three freight transportation



(a) (b)

Fig. 1 Carbon emissions and Freight turnover in 2016. **a** Carbon emissions in 2016. **b** Freight turnover in 2016

modes. Then, with the goal of maximizing the efficiency of the freight transportation system, considering the carbon emission intensity reduction constraint and the economic output non-reduction constraint, this paper constructs a freight structure adjustment model. Finally, the freight structure adjustment plan in 2020 was calculated and some policy suggestions are given.

This paper mainly has the following two contributions. (1) We calculated and analyzed the characteristics of the economic benefits, carbon emission intensity, and input-output efficiency of different freight modes in different provinces, and help decision-makers in the freight market to further understand the characteristics of different freight modes in different provinces. (2) We have built a freight structure adjustment model to optimize the efficiency of the freight transport structure and meet the constraints of economic output growth and carbon emission intensity decline, so that the freight decision-makers of each province can further clarify how the freight structure should be adjusted to meet the emission reduction requirements and optimize the efficiency of the freight transportation system.

The rest of the paper is structured as follows. "Methodology" introduces the various research methods and data used. In "Results," the empirical results and discussions are provided. Finally, the conclusions, policy implications, and some research issues to address in the future are presented in "Conclusions."

Methodology

The freight structure adjustment model

The freight structure adjustment model design is shown in Fig. 2. We use the panel data model to quantify the economic output of different freight modes, that is, the economic output per unit of freight turnover; use the carbon emission calculation formula to calculate the carbon emissions per unit freight turnover and carbon emission intensity, which is the carbon emissions per unit output value; and use the DEA-SBM model to calculate the input-output efficiency, that is, the output of freight turnover under the same input. Finally, we built a freight structure adjustment model to maximize the efficiency of the freight transportation system while meeting the constraints of reducing carbon emission intensity and increasing economic output.

With the goal of maximizing the efficiency of the freight transportation system and meeting the constraints of increasing economic output and reducing carbon emission intensity, the freight structure adjustment model is established as follows:

$$\max\ Z = \alpha_i X_i + \beta_i Y_i \tag{1}$$

$$\text{s.t.}\ \ (a_i - b_i)X_i + (c_i - b_i)Y_i \geq 0 \tag{2}$$

$$\frac{(RT_i^m + X_i)*ef_r + (HT_i^m - X_i - Y_i)*ef_h + (WT_i^m + Y_i)*ef_w}{RT_i^m + HT_i^m + WT_i^m} \leq C_i^n*(1-w_i) \tag{3}$$

$$\frac{(RT_i^m + X_i)*ef_r + (HT_i^m - X_i - Y_i)*ef_h + (WT_i^m + Y_i)*ef_w}{(RT_i^m + X_i)*a_i + (HT_i^m - X_i - Y_i)*b_i + (WT_i^m + Y_i)*c_i} \leq CI_i^n*(1-\Delta CI_i) \tag{4}$$

$$X_i + Y_i \leq 0.5*HT_i^m \tag{5}$$

$X_i \geq 0,\ Y_i \geq 0;\ i = 1, 2$,where:

i	represents each province.
X_i and Y_i	represent freight volume transferred from roadway to railway, and from roadway to waterway respectively, which are our targets.
α_i and β_i	are the relatively insufficient output value of roadway compared to railway and waterway respectively.

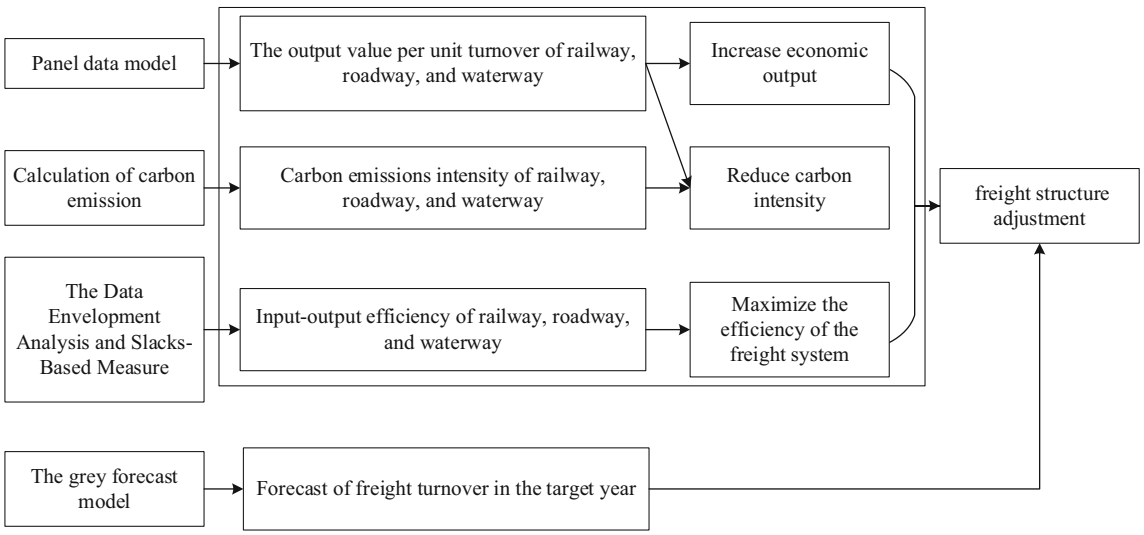


Fig. 2 The freight structure adjustment model design

a_i, b_i, and c_i are the output value per unit turnover of railway, roadway, and waterway respectively.

ef_r, ef_h, and ef_w are carbon emissions per unit turnover of railway, roadway, and waterway respectively (10,000 tons CO_2/100 million ton-kilometers).

RT_i^m, HT_i^m, and WT_i^m are the forecasting freight turnover volume of railway, roadway, and waterway in the target year (100 million ton-kilometers).

C_i^n represents the carbon emissions generated by per-unit freight turnover in the reference year (10,000 tons CO_2/100 million ton-kilometers).

w_i represents the specific reduction requirement of carbon emission in each province (compared with the reference year).

CI_i^n represents the carbon emissions generated by the unit output value (10,000 tons of CO_2/100 million yuan).

ΔCI_i is the specific reduction requirement of carbon emission intensity in each province (compared with the reference year).

The carbon emissions generated by per-unit freight turnover in reference year:

$$C_i^n = \frac{CO_{2_i}^n}{RT_i^n + HT_i^n + WT_i^n + AT_i^n} \tag{6}$$

The carbon emissions generated by the unit output value in reference year:

$$CI_i^n = \frac{CO_{2_i}^n}{a_i{*}RT_i^n + b_i{*}HT_i^n + c_i{*}WT_i^n} \tag{7}$$

The objective function indicates that under the same input, the increased output of the transportation system after the road freight turnover is converted to the railway and the waterway by X_i and Y_i respectively. Constraint (2) indicates that the output value of the freight transport system after redistribution should not be less than that before adjustment. Constraint (3) indicates that the carbon emissions per unit turnover of freight transportation system should be less than the set threshold. Constraint (4) indicates that the carbon emissions generated by the unit output value in each province should meet the decline requirements. Constraint (5) means that the sum of the road freight turnover is converted to the railway and the waterway does not exceed 50% of the road freight turnover in the target year. According to the "Preliminary Investigation and Research on Freight Industry" study on the proportion of China's road freight transportation within and across provinces (STL, SWJTU 2020), China's inter-provincial transportation accounted for more than 50% in 2017. Considering that the distance of transportation of goods within the province is generally short, it is not suitable for railway and waterway; therefore, this paper set the transferable proportion threshold not to exceed 50% of the total share of road freight.

Panel data model

Panel data analysis is a statistical analysis method for analyzing two-dimensional data (cross-sectional and longitudinal), and in the spatiotemporal dimension, it has many advantages over simple regression analysis. For example, the panel unit root test can test the stability of variables, avoiding spurious regression. The long-time equilibrium relationship among variables can be determined by panel co-integration test. The panel data model also considers more sample variation and more degree of freedom to improve the prediction efficiency.

When researching the relationship between transportation and economy, some scholars applied panel data analysis to test the relationship between transport and economy (Beyzatlar et al. 2014). In this paper, the panel data model is adopted to analyze the impact of three transport modes (railway, roadway, and waterway) on economy; the degree of impact is measured by the output value of unit freight turnover in different provinces.

Specific coefficient model of section members is used in this paper to study the unit freight turnover economic output value of railway, roadway, and waterway. The panel models are constructed as follows:

$$Y_{it} = \alpha_i + a_i RT_{it} + bHT_{it} + cWT_{it} + \varepsilon_{it} \tag{8}$$

$$Y_{it} = \alpha_i + aRT_{it} + b_i HT_{it} + cWT_{it} + \varepsilon_{it} \tag{9}$$

$$Y_{it} = \alpha_i + aRT_{it} + bHT_{it} + c_i WT_{it} + \varepsilon_{it} \tag{10}$$

In which, Y_{it} is the output value of the transportation industry in province i of year t. RT_{it}, HT_{it}, and WT_{it} represent the converted turnover of railway, roadway, and waterway in province i of year t. α_i is the constant term. ε_{it} represents residual. a_i, b_i, and c_i are the specific coefficients. Formula (8), (9), and (10) take RT_{it}, HT_{it}, and WT_{it} as their specific variables respectively, and for different provinces, the coefficient of this variable is different, while coefficients for other variables do not vary with the provinces. This method can be used to calculate the impact of the change of one transport mode turnover on the output value of the transportation industry under the condition that other factors remain unchanged.

In particular, the converted turnover of the three transportation modes = freight turnover + (passenger turnover × coefficient of passenger turnover converted into freight turnover). The passenger and freight conversion factors are shown in Table 1.

Table 1 Coefficient of passenger turnover converted into freight turnover

Railway	Roadway	Waterway
1	0.1	0.33

Data source: "China Railway Yearbook" and "China Traffic and Transportation Statistical Yearbook"

Calculation of carbon emission

Now, there are two calculation methods for carbon emission in the transportation industry: top-down and bottom-up. The top-down method (Han et al. 2017; IPCC 2014; Zhou et al. 2013; Guo et al. 2014; Xu and Lin 2016) is based on the statistical data of fuel consumption, whereas the bottom-up method is based on the vehicle ownership, mileage, fuel consumption per mileage, and carbon emission coefficient of various fuels (Tongwane et al. 2015; Tian et al. 2014; To 2015; Hao et al. 2015b). Considering the availability of data, the top-down method is more popular in China and this paper will adopt the top-down method. According to the method provided in IPCC AR5 (IPCC 2014), the formula is as follows:

$$E_i = \sum_j T_{ij} EF_{ij} C_j \tag{11}$$

where i denotes the transportation modes (railway, roadway, waterway). j denotes fuel type. E_i is the carbon emission of transportation mode i. T_{ij} indicates the freight turnover of transportation mode i using fuel j (100 million ton-kilometers). EF_{ij} is unit turnover energy consumption by transportation mode i using fuel j. C_j is carbon emission coefficient of fuel j.

Input-output efficiency of different modes of transport

DEA is a linear programming method to measure the efficiency of multiple decision-making units, where there are multiple inputs and outputs in the production process. The data envelopment analysis method proposed by Charnes et al. (1978) applied to the non-parametric mathematical programming of frontier estimation, namely the CCR model, is the most basic and important technology in the DEA model. The principle is to keep the input or output of decision-making units (DMU) unchanged, use mathematical planning and statistical data to determine a relatively effective production frontier, project each decision unit onto the production frontier of DEA, and compare the degree to which the decision unit deviates from the frontier of DEA; to evaluate its relative effectiveness is suitable for the comprehensive evaluation of the effectiveness of multiple outputs and multiple inputs. A large number of studies have followed, considering a series of other assumptions, to expand the DEA method. Tone (2001) proposed the SBM model. Compared with the CCR model, the SBM model mainly considers the maximization of profit rather than the maximization of the input-output ratio, and the efficiency score of the SBM model evaluated during efficiency evaluation is less than the CCR evaluation score value, which is convenient for comparison and sorting of various decision-making units. This paper uses the DEA-SBM model to evaluate the efficiency of input-output efficiency of railway, roadway, and waterway. The principles of the SBM model are as follows:

Supposing there are N decision-making units, and each decision-making unit has M inputs and S outputs, the input matrix and out matrix are expressed by $X = (x_{ij}) \in R^{m \times n}$ 和 $Y = (y_{ij}) \in R^{s \times n}$ respectively. (Both input matrix and output matrix are non-negative).

Define possible production set: P = {(x, y)|x > = Xλ, y < = Xλ, λ > = 0}

For a certain decision-making unit (x_o, y_o), it can be expressed as:

$$x_o = X\lambda + s^- \tag{12}$$

$$y_o = Y\lambda - s^+ \tag{13}$$

In which, $\lambda \in R^n$, and $\lambda \geq 0$; s^-, s^+ denote input redundancy and output shortage respectively, also called slack variables, and $s^- \in R^m$, $s^+ \in R^s$, s^-, $s^+ \geq 0$.

Then, DEA-SBM model can be built as follows:

$$\min \ \rho = \frac{1 - 1/m\sum_{j=1}^{m} s_j^- / x_{io}}{1 + 1/s\sum_{r=1}^{s} s_r^+ / y_{ro}} \tag{14}$$

Table 2 Input-output variables

Variable	Railway	Roadway	Waterway	Unit	Type
Fixed-asset investment	√	√	√	Billion	Capital
Practitioners	√	√	√	People	Labor
Diesel	√	√	√	L	Energy
Gasoline	√	√	-	L	Energy
Electricity	√	√	√	kWh	Energy
Cargo turnover	√	√	√	Billion ton-kilometers	Output

Table 3 Descriptions on data

Item	Number of provinces	Period	Source
The output value of the transportation industry	31	2000–2017	The statistical yearbook of each province.
Cargo turnover	31	2000–2017	The statistical yearbook of each province.
Passenger turnover	31	2000–2017	The statistical yearbook of each province.
Passenger and cargo conversion factor	-	2017	"China Railway Yearbook" "China Transport Statistical Yearbook",
Energy consumption per turnover by fuel type	-	2016	"China Transportation Yearbook", "China Logistics Yearbook", "China Railway Yearbook"
The carbon emission factor of different energy type	-	2013	World Resources Institute "Guidelines for the Calculation of Greenhouse Gas Emissions from Energy Consumption Tools Version 2.1"
Fixed capital investment of transport mode	31	2006–2016	"China Transport Statistical Yearbook", "China Railway Statistical Yearbook"
Practitioners by mode of transport	31	2006–2016	"China Transport Statistical Yearbook", "China Railway Statistical Yearbook"

$$\text{s.t. } x_o = X\lambda + s^- \quad (15)$$

$$y_o = Y\lambda - s^+ \quad (16)$$

$$\lambda \geq 0, s^- \geq 0, \ s^+ \geq 0$$

The interval of the objective function ρ is [0,1]; when ρ = 1 and $s^-, s^+ = 0$, the decision-making unit is effective. When ρ < 1 and $s^-, s^+ \neq 0$, the decision-making unit is ineffective, so the input and output need to be improved. From formula (16), we can get the insufficient output value s^+ of roadway, railway, and waterway respectively under the same input. After normalization and subtraction, we get the relative insufficient output value α_i , β_i of roadway compared to railway and waterway respectively.

We referred to Li and Lin (2016), Li and Wu (2017), and Sun et al. (2020a) in selecting the input and output variables. We select capital, labor, and energy as input factors. Capital is represented by fixed-asset investment, and labor is represented by employees and energy includes diesel and gasoline as well as electricity. The output is measured by the freight turnover of three transportation modes (Table 2).

Data source: "China Transport Statistical Yearbook"; "China Railway Statistical Yearbook 2017"

Forecasting method of freight turnover

When predicting the freight turnover in the target year, it can be predicted by a variety of methods based on historical data. For the sake of simplicity, this paper chooses the gray time series forecasting model. At present, there are many studies that prove that the gray time series forecasting model is an effective method that can use even small data to predict the future patterns of a phenomenon (Liu et al. 2020; Tsai et al. 2017; Hamzacebi and Es 2014; Lim et al. 2010). The gray forecast model predicts the freight turnover of the future period based on the freight turnover of the known period. The solution of the gray prediction model consists of three steps. Gray model (GM model for short), GM (1,1) means to use a first-order differential equation to model one variable which is the freight turnover.

1) Data testing

Table 4 Carbon emission intensity reduction requirements

Province	2020 carbon emission intensity reduction requirements compared to 2015 (reference year)
Beijing, Tianjin, Hebei, Shanghai, Jiangsu, Zhejiang, Shandong, Guangdong	20.5%
Fujian, Jiangxi, Henan, Hubei, Chongqing, Sichuan	19.5%
Shanxi, Liaoning, Jilin, Anhui, Hunan, Guizhou, Yunnan, Shaanxi	18.0%
Inner Mongolia, Heilongjiang, Guangxi, Gansu, Ningxia	17.0%
Hainan, Tibet, Qinghai, Xinjiang	12.0%

Table 5 Results of unit root test

		Y	RT	HT	WT
LLC t-value	Level	12.148	−3.687***	6.422	5.483
	First-order difference	−9.480***	−9.150***	−15.882***	−13.869***
IPS Wvalue	Level	17.648	1.888	9.0423	6.455
	First-order difference	−6.931***	−8.860***	−12.155***	−10.309***
ADF-Fisher Chi-square	Level	3.819	46.946	27.926	28.841
	First-order difference	162.768***	189.222***	253.155***	222.460***
PP-Fisher Chi-square	Level	4.154	44.067	9.006	23.800
	First-order difference	204.158***	169.014***	257.021***	334.239***

Note: * represent significance at the level of 10%. ** represent significance at the level of 5%. *** represent significant at the level of 1%

To ensure the feasibility of the GM (1,1) forecasting method, it is necessary to test the data.

Assuming that the original data is listed as $x^{(0)} = (x^{(0)}(1), x^{(0)}(2), \ldots, x^{(0)}(n))$, then calculate the ratio of a sequence:

$$\lambda(\mathrm{k}) = \frac{x^{(0)}(k-1)}{x^{(0)}(k)}, \mathrm{k} = 2, 3, \ldots, \mathrm{n} \tag{17}$$

If all ratios fall within the capable coverage range $\mathrm{X} = \left(e^{\frac{-2}{n+1}}, e^{\frac{2}{n+1}}\right)$, then GM(1,1) can be used to forecast sequence $x^{(0)}$. Otherwise, the origin data need to be properly transformed, such as translation transformation:

$$y^{(0)}(k) = x^{(0)}(k) + c, \mathrm{k} = 1, 2, \ldots, \mathrm{n} \tag{18}$$

Choose the appropriate value c, so that the ratio of the sequence falls within the capable coverage.

2) Establishment of GM(1,1) prediction model

Let the sequence $x^{(0)} = (x^{(0)}(1), x^{(0)}(2), \ldots, x^{(0)}(n))$ satisfy the condition: all ratios fall within the capable coverage range $\mathrm{X} = \left(e^{\frac{-2}{n+1}}, e^{\frac{2}{n+1}}\right)$, so it can be used to build GM(1,1) model:

$$\mathrm{x}^{(0)}(k) + \alpha z^{(1)}(k) = b \tag{19}$$

The estimates of α and b are obtained by the least square method, and the corresponding whitening model is established:

$$\frac{dx^{(1)}(t)}{dt} + \alpha x^{(1)}(t) = b \tag{20}$$

The solution is:

$$x^{(1)}(t) = \left(x^{(0)}(1) - \frac{b}{a}\right)e^{-a(t-1)} + \frac{b}{a} \tag{21}$$

Then the prediction value of $\widehat{x}^{(1)}(k+1)$ is obtained:

$$\widehat{x}^{(1)}(k+1) = \left(x^{(0)}(1) - \frac{b}{a}\right)e^{-ak} + \frac{b}{a}, k = 1, 2, \ldots n-1 \tag{22}$$

So, the prediction value of $\widehat{x}^{(0)}(k+1)$ is obtained:

$$\widehat{x}^{(0)}(k+1) = \widehat{x}^{(1)}(k+1) - \widehat{x}^{(1)}(k), k = 1, 2, \ldots n-1 \tag{23}$$

3) Residual testing

Computing relative residual

$$\varepsilon(\mathrm{k}) = \frac{x^{(0)}(k) - \widehat{x}^{(0)}(k)}{x^{(0)}(k)}, \quad k = 1, 2, \ldots n \tag{24}$$

If all $|\varepsilon(\mathrm{k})| < 0.1$, it is considered that the prediction is accurate. If all $|\varepsilon(\mathrm{k})| < 0.2$, it is considered that the prediction is generally accurate.

Table 6 Kao cointegration test result

ADF	t-Statistic	Prob.
	3.135	0.001***
Residual variance	9187.713	
HAC variance	14586.500	

Data

Through the published "China Transportation Yearbook," "China Logistics Yearbook," "China Railway Yearbook," and "China Transport Statistical Yearbook," and the national and local government statistical bureaus, we have obtained the output value of the transportation industry, the cargo turnover volume of three transportation modes, passenger turnover volume, coefficient of passenger turnover converted into cargo turnover, energy consumption per unit turnover volume of different fuel types, CO_2 emission coefficients corresponding

Table 7 Johansen cointegration test result

Hypothesized No. of CE(s)	Fisher Stat.* (from trace test)	Prob.	Fisher Stat.* (from max-eigen test)	Prob.
None	1153.0	0.000	835.2	0.000***
At most 1	574.3	0.000	438.3	0.000***
At most 2	222.0	0.000	158.2	0.000***
At most 3	117.9	0.000	106.9	0.000***
At most 4	76.2	0.037	76.24	0.037**

to various energy types, and fixed capital investment and employees of three transportation modes. For details, see Table 3.

The "13th Five-Year Plan for Controlling Greenhouse Gas Emissions" proposed the specific reduction requirement of carbon emission intensity (carbon emissions per unit of output value) in each province shown in Table 4.

Results

Panel unit root test

Before conducting the panel analysis, the unit root test and the cointegration test must be performed first. The unit root test (Table 5) shows that all sequences are first-order stationery. Therefore, it is feasible to use these variables for the cointegration test.

The cointegration test is mainly used to test whether there is a long-term equilibrium relationship between dependent variables and independent variables (Sun et al. 2020c). The commonly used cointegration test methods include Kao test, Johansen test, and Pedroni test. Gutierrez (2003) used Monte Carlo simulation to compare several methods of cointegration test, showing that Kao test is more effective than Pedroni test when t value is small. So Kao test and Johansen test are used to doing the cointegration test in this paper. The result of Kao test is shown in Table 6 and the result of Johansen test is shown in Table 7. It can be seen that the Kao test and the Johansen test rejected the null hypothesis that there is no cointegration relationship in all regions, indicating that the transportation industry output value and the freight turnover volume maintain a long-term cointegration relationship during the study period.

Panel data analysis includes three models: mixed model (MM), fixed effects model (FEM), and random effects model (REM). In order to adopt a suitable model, it needs to test the regression results of the three models. As is shown in Table 8, the fixed impact model test rejected the original hypothesis at a significant level of 0.1%; hence, the fixed impact model is chosen in this paper.

The results of panel data analysis are shown in Fig. 3 we can see that, for the three transportation modes, the economic output value of unit turnover in northwest China is relatively lower, which in the eastern region is higher; this is mainly because the output value per unit turnover is closely related to the value of goods, and the economic development level of the southeast coastal area is high. In almost all provinces, compared with railway and waterway, the unit turnover output value of roadway is the lowest. For railway freight transportation, the production value per unit turnover has great regional differences. The provinces in the southeast coastal areas such as Shanghai, Jiangsu, and Fujian have high unit turnover volume output value. The unit turnover volume output value of railways in these provinces all exceeded 300 yuan/100 ton-kilometers. For road transportation, the economic output per unit turnover volume of Beijing and Tianjin is relatively high; the unit turnover volume output value exceeds 120 yuan/100 ton-kilometers. For water transportation, Jiangsu, Zhejiang, Guangdong, and Fujian have higher unit turnover volume output value, and the economic output per unit turnover volume is greater than 500 yuan/100 ton-kilometers.

Table 8 Fixed impact model test

Model	Effects test	Statistic	d.f.	Prob.
The panel model (1)	Cross-section F	6.918***	(30,487)	0.000***
	Cross-section Chi-square	195.940***	30	0.000***
The panel model (2)	Cross-section F	7.518***	(30,487)	0.000***
	Cross-section Chi-square	210.082***	30	0.000***
The panel model (3)	Cross-section F	14.038***	(27445)	0.000***
	Cross-section Chi-square	310.521***	27	0.000***



Fig. 3 Output value per unit freight turnover of three transportation modes in different regions

Carbon emission calculation

According to China's 13th Five-Year Plan, the task of reducing carbon emissions for the transportation industry in 2020 is based on the percentage reduction of carbon emissions in 2015; therefore, we need to calculate carbon emissions in 2015. According to Formula11, we obtained the carbon emissions of the transportation industry in each province in 2015. Figure 4 shows the total freight carbon emissions of various provinces in 2015. The bluer the color, the greater the carbon emissions. In 2015, almost all provinces have the largest carbon emissions from road transportation, and only a few

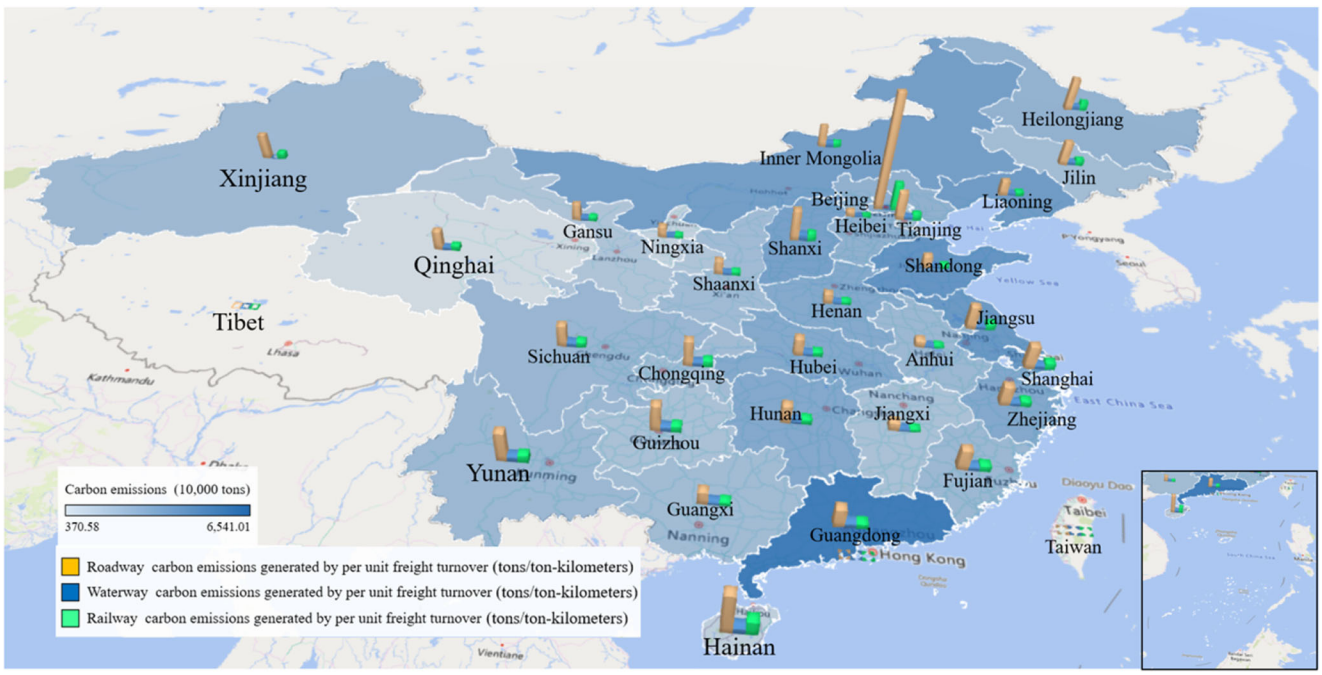


Fig. 4 Carbon emissions of three transportation modes in different provinces

coastal provinces have more carbon emissions from water transportation than road transportation. The carbon emissions in eastern China were significantly higher than those in the western regions; Guangdong, Shandong, and Shanghai had the highest carbon emissions. The bar chart in Fig. 4 shows the carbon emissions generated per unit freight turnover of each transportation mode. Road transportation has the largest carbon emission per unit freight volume, followed by railway transportation, and water transportation has the smallest carbon emission per unit freight volume.

Input-output efficiency calculation

DEA-SBM model is used to measure the average input-output efficiency of railway, roadway, and water transport in 31 provinces during 2001 to 2016. Selecting capital, labor, and energy as input factors, capital is represented by fixed-asset investment, and labor is represented by employees and energy includes kerosene, diesel, and gasoline as well as electricity. The output is measured by the freight turnover of three transportation modes. Table 9 shows the input-output efficiency of railway, roadway, and water transport in 31 provinces.

The waterway transportation in the coastal areas such as Liaoning, Shandong, Jiangsu, Shanghai, Zhejiang, Fujian, Guangdong, Hainan, and the inland provinces adjacent to the Yangtze River shows a higher input-output efficiency compared with other transportation modes in the province. Among them, Shanghai has the highest input-output efficiency of 0.8106, and Zhejiang Province ranks second, with an input-output efficiency of 0.7690. The central and western regions depend mainly on road and rail transportation. In inland regions, except for provinces along the Yangtze River,

Table 9 Input-output efficiency of three transportation modes

Provinces	Railway relative efficiency	Highway relative efficiency	Water transport relative efficiency
Beijing	0.8421	0.2547	-
Tianjin	0.1082	0.0829	0.4673
Hebei	0.3386	0.2651	0.4910
Shanxi	0.8537	0.7108	-
Inner Mongolia	0.7703	0.5479	-
Liaoning	0.1294	0.1347	0.7238
Jilin	0.7239	0.6930	0.1137
Heilongjiang	0.7843	0.6378	0.1490
Shanghai	0.0709	0.1325	0.8106
Jiangsu	0.3025	0.0619	0.6693
Zhejiang	0.1379	0.0526	0.7690
Anhui	0.1927	0.0982	0.4705
Fujian	0.2903	0.0528	0.6133
Jiangxi	0.8324	0.3905	0.6632
Shandong	0.2549	0.2611	0.5799
Henan	0.4261	0.0854	0.6285
Hubei	0.5094	0.0499	0.5645
Hunan	0.8396	0.1230	0.5997
Guangdong	0.2357	0.1792	0.4276
Guangxi	0.6428	0.1392	0.5733
Hainan	0.0387	0.0594	0.5228
Chongqing	0.1501	0.0300	0.6136
Sichuan	0.8871	0.2756	0.4496
Guizhou	0.8262	0.1988	0.4186
Yunnan	0.8749	0.5005	0.5474
Tibet	0.5595	0.0238	-
Shaanxi	0.8265	0.4751	0.1047
Gansu	0.8119	0.2896	-
Qinghai	0.7726	0.5571	-
Ningxia	0.8447	0.6894	-
Xinjiang	0.7948	0.5132	-

the input-output efficiency of rail transportation mode in other provinces is generally higher, and its range is roughly between 0.7 and 0.9. In most provinces in the western region, the input-output efficiency of railway is greater than 0.8. For most provinces, the input-output efficiency of roadway is lower than that of railway and waterway. Generally, the input-output efficiency of waterway in coastal areas and provinces adjacent to the Yangtze River is higher, while in inland regions, the railway is more efficient.

From formula (16), we can get the insufficient output value s^+ of roadway, railway, and waterway respectively in each province. After normalization and subtraction, we get the relative insufficient output value α_i, β_i of roadway compared to railway and waterway respectively, as shown in Fig. 5. The insufficient output of roadways in almost all provinces is greater than that of railways and waterways.

Discussion on the provincial freight structure shift

After solving the freight structure adjustment model with the CO_2 emissions saving target proposed in the 13th Five-Year Plan for the Development of Modern Integrated Transport System, we could get the freight structure shift result in Table 10, in which both the detailed shift turnover volumes from roadway to railway and roadway to waterway at a province-level are given.

With the maintaining of the current level of economic outputs from the freight sector, only some provinces are still with the potential of shifting the freight turnover from the roadway to other transportation modes, which are presented in Table 10. In other words, in the other provinces or regions in China, the feasible measures for reducing the carbon emissions related to the freight transportation sector are decoupling of economic development; as well as improvement of the emission efficiency of road freight.

Table 10 also gives the ranks according to the freight turnover volume that should be transferred from road transportation to other transportation modes at the provincial level. Tianjin, Anhui, Beijing, Hebei, and Jiangxi rank 1 to 5. The provinces or regions that ranked 6–8 are Tibet, Heilongjiang, and Shandong, whose shift proportion is more than 5%. Ningxia, Qinghai, Jilin, Xinjiang, Shaanxi, and Gansu are ranked following, all of which have less room for roadway transportation transfer.

It is with great potential for Tianjin to shift its freight turnover from the roadway to other transport modes without sacrificing its economic benefits, which is 0.21% and 18.65% from the roadway to railway, and roadway to the waterway, respectively. It is worth noting that, as an important port city in North China, Tianjin still seldom uses its powerful water transport capacity.

Anhui is an important province in the middle and lower reaches of the Yangtze River. It already has good water transportation conditions. It has realized the water transportation of goods between cities in the province, but the railway transportation capacity still has great potential. Therefore, Anhui is also with potential to shift its freight volume and ranks at no.2, with 15.36% transfer to the railway, as well as 0.23% transfer to the waterway.

The railway capacity of Beijing can be further stimulated; according to the results, 14.12% of road freight should be transferred to railways.

For Hebei province, similar to Tianjin, the potential of roadway transfer to the waterway is larger than that to the railway; however, the difference between the two modes of transportation is smaller. The freight that could be converted

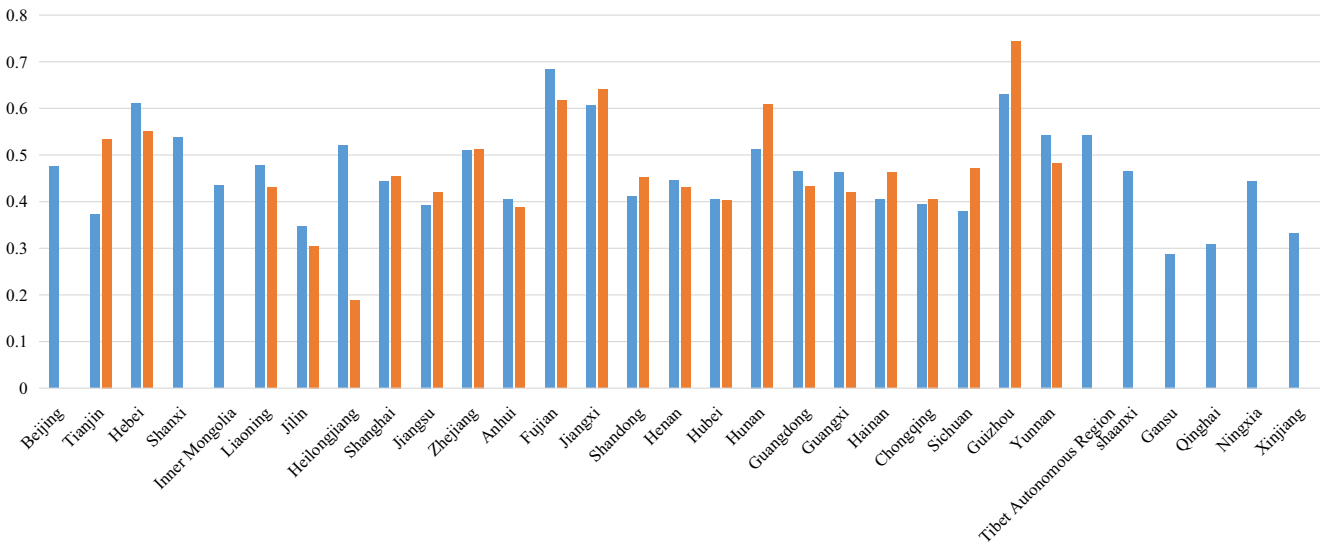


Fig. 5 The inefficient of roadway compared to railway and waterway

Table 10 The conversion turnover volume from roadway to railway and waterway

Province	Roadway → railway (100 million tons.km)	The convert proportion	Roadway → waterway (100 million tons.km)	The convert proportion	The total convert proportion of roadway	Rank
Beijing	26	14.12%	-	-	14.12%	3
Tianjin	1	0.21%	87	18.44%	18.65%	1
Hebei	372	3.94%	746	7.90%	11.84%	4
Jilin	6	0.45%	13	0.99%	1.44%	11
Heilongjiang	70	7.36%	2	0.21%	7.57%	7
Anhui	658	15.36%	10	0.23%	15.59%	2
Jiangxi	430	10.21%	28	0.66%	10.87%	5
Shandong	5	0.08%	387	6.45%	6.53%	8
Tibet	16	8.22%	-	-	8.22%	6
Shaanxi	2	0.08%	-		0.08%	13
Gansu	1	0.08%	-		0.08%	13*
Qinghai	4	1.74%	-		1.74%	10
Ningxia	16	3.35%	-		3.35%	9
Xinjiang	13	0.79%	-		0.79%	12

from roadway to waterway is related to the port cities, such as Tangshan, Huangye, and Qinhuangdao.

The freight transfer to railway and waterway almost equal for Jiangxi indicates that Jiangxi should make use of the capacity both for railway and waterway.

The Qinghai-Tibet Railway and the Sichuan-Tibet Railway under construction have increased railway transportation capacity in Tibet. In the future, Tibet could consider transferring road transportation to railways.

Overall, the regional governments of Anhui, Beijing, Jiangxi, Tibet Autonomous Region, and Heilongjiang should focus on transferring the freight share of roadway to the railway, and the transferred shares are 15.36%, 14.12%, 10.21%, 8.22%, and 7.36% of road freight turnover. The governments of Tianjin, Hebei, and Shandong should focus on transferring the freight share of roadway to the waterway, and the transferred shares are 18.44%, 7.90%, and 6.45% of road freight turnover.

Conclusion

This paper studies and analyzes the economic output, carbon emission intensity, and input-output efficiency of the three freight transportation modes at the provincial level in China. By constructing a panel analysis model, this paper quantitatively analyzes the economic output value of different freight transportation modes. It is found that waterway is the transportation mode with the highest output value per unit turnover, followed by railway, and the lowest is roadway in most provinces. The output value of unit turnover of the same transportation mode also varies greatly in different regions; the economic output value of unit turnover in northwest China is relatively lower, which in the eastern region is higher. Through the carbon emission calculation formula, the carbon emissions intensity of different freight modes is clarified. Road transportation has the largest carbon emission per unit freight volume, followed by railway transportation, and water transportation has the smallest carbon emission per unit freight volume. The carbon emissions in eastern China were significantly higher than those in the western regions. The DEA-SBM model is used to quantify the difference in input-output efficiency of the three freight modes, railway and waterway are more efficient, and roadway efficiency is the lowest. The waterway output efficiency is highest for most eastern provinces, and for most of the provinces in central China, the output efficiency of railway is the highest.

With the goal of maximizing the efficiency of the freight transportation system, while meeting the constraints of increasing economic output and reducing carbon emissions, this paper establishes a freight structure adjustment model. Based on the optimization model proposed in this paper, under the constraint of the energy conservation and emission reduction targets by reducing about 7% proportions of carbon emissions compared with the year 2015, the specific volume of roadway turnover transferred to waterway and railway in each province is given. The results show that the regional governments of Anhui, Beijing, Jiangxi, Tibet Autonomous Region, and Heilongjiang should focus on transferring the freight share of roadway to railway, and the transferred shares are

15.36%, 14.12%, 10.21%, 8.22%, and 7.36% of road freight turnover; the governments of Tianjin, Hebei, and Shandong should focus on transferring the freight share of roadway to waterway, and the transferred shares are 18.44%, 7.90%, and 6.45% of road freight turnover, providing a reference for relevant government departments to formulate measures.

Despite these findings of this research, there are still some limitations that should be pointed out. First, due to the real case study data availability, such as the mandatory reducing value of emissions, this research only applied the emission reduction targets set by China in the 13th Five-Year Plan; however, the methodology and models are general; the Practitioner could apply our model to formulate more timely adjustment plans based on the target of 14th Five year plan. Second, although the road freight is far from energy-efficient, it preserves the features of convenience and flexibility, and cannot be replaced by other freight modes, especially in the situation of short-distance transport. Thus, allocating the turnover volume share cannot solely rely on the economic benefit and carbon emission constraint. In the future study, we will apply the OD data of road freight, to analyze the lower bound share of road freight turnover for each province of China, and propose a more realistic model. Third, this paper uses a relatively macro model to calculate the carbon emission intensity and economics of the freight model. In the future, scholars can continue to optimize the carbon emission intensity and economic calculation methods of the freight model to make it more realistic and to formulate more precise adjustment plans.

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Declarations

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